MICHAEL L. LOPEZ

Senior DevOps Engineer | Platform Engineering & Cloud Infrastructure

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PROFESSIONAL SUMMARY

Senior DevOps Engineer with 15 years across cloud infrastructure, platform engineering, and production operations. Specializes in building internal developer platforms — Terraform modules, GitOps workflows via ArgoCD, and Kubernetes-native tooling that engineering teams adopt at scale. Extended platform work into AI-augmented engineering: multi-agent code review, governance frameworks for production and IAM operations, and context engineering across 6+ infrastructure repositories. AWS practitioner with hands-on expertise across EKS, Confluent, CI/CD automation, and event-streaming platforms.

CORE COMPETENCIES

Cloud & Infrastructure: AWS, Terraform, Docker, Kubernetes

CI/CD & GitOps: ArgoCD, Concourse CI, Jenkins, Helm

Event Streaming & Data: AWS MSK, Confluent for Kubernetes, Confluent Cloud, Apache Kafka

Observability & SRE: Sumo Logic, New Relic, Grafana, PagerDuty, AWS CloudWatch

Languages & Scripting: Python, Shell/Bash, SQL, YAML, Java

AI Platform Engineering: Multi-Agent Orchestration, Context Engineering, Safety Governance, Memory Schema Design, MCP Integrations (Jira/GitHub/New Relic/Docker), AI-Assisted IaC, Agentic Code Review

PROFESSIONAL EXPERIENCE

Senior DevOps Engineer | Trimble | Remote *Sep 2024 – Present*

Lead platform engineering for a multi-team cloud-native environment, owning the full lifecycle of internal developer tooling and infrastructure services.

- Own and operate the full platform stack across multiple AWS accounts and environments (development, staging, and production): EKS cluster lifecycle and version upgrades (v1.24 through v1.33), ArgoCD-driven GitOps deployments, Concourse CI, Confluent for Kubernetes upgrades, and Confluent Cloud — reducing deployment time from 1 day to 10 minutes and MTTR from 6 hours to 5 minutes.
- Manage secrets infrastructure at scale using AWS Secrets Manager and External Secrets Operator (ESO), consolidating IAM policies via Terraform and maintaining secrets hygiene across 20+ services spanning development, staging, and production accounts.
- Led full TLS/SSL certificate migration for all Kubernetes load balancers across four separate AWS accounts, coordinating phased validation and cutover with zero service disruption.
- Own platform security posture: CrowdStrike endpoint agent rollout via ArgoCD daemonsets, FIPS 140-2 enablement on Confluent for Kubernetes in the GovCloud environment, Log4j CVE remediation across all compute instances, Snyk vulnerability remediation, and internal cybersecurity compliance validation.
- Deployed KEDA (Kubernetes Event-Driven Autoscaling) across all clusters via ArgoCD, enabling workloads to scale automatically based on event queue depth; built full Concourse CI observability stack including Prometheus metrics, RDS Enhanced Monitoring, and New Relic dashboards.
- Engineered reusable Terraform modules for AWS infrastructure provisioning, cutting infrastructure provisioning time from 1 day to 30 minutes and eliminating configuration drift across environments.
- Leading migration of all Kafka workloads from Confluent Platform (CFK + Confluent Cloud) to AWS MSK — authored the MSK Terraform module, provisioned clusters with mTLS across GovCloud and commercial environments, and coordinating phased cutover across 15+ environments.

- Modernized Concourse CI infrastructure: migrated databases to dedicated RDS PostgreSQL instances (PostgreSQL 11 to 17 upgrade via Terraform), upgraded to latest stable versions across all environments, and built FIPS-compliant container images.
- Manage Kong API Gateway lifecycle across all environments, including version upgrades (3.4 to 3.10) with dedicated nodegroup taints and tolerations for workload isolation.
- **Architected a multi-agent code review platform** for infrastructure workflows — orchestration layer routes PRs to domain-specific agents for Terraform and ArgoCD, reducing reviewer load on IaC changes and catching drift before human review; deployed as internal team tooling with documentation and adoption materials.
- **Designed AI governance and context engineering framework** across 6+ infrastructure repositories — eliminated unreviewed production and IAM changes through enforced approval gates; shortened agent onboarding time per new repo through standardized context and knowledge schemas.

DevOps Engineer | Trimble | Remote *Jul 2019 – Sep 2024*

Grew from individual contributor to platform owner over 5 years, driving the team's shift from manual deployments to fully automated, GitOps-driven infrastructure.

- Provisioned and administered Concourse CI and Jenkins infrastructure, establishing reusable pipeline templates (build, test, multi-environment promotion) that development teams adopted as the standard for Kubernetes-based deployments across the organization.
- Drove Kubernetes adoption across engineering, provisioning cluster infrastructure and deploying Helm charts for platform services including Kong API gateway, Confluent, ingress controllers, and monitoring tooling — supporting the migration of 15–20 services from virtual machines to EKS.
- Pioneered Terraform IaC adoption, authoring AWS provisioning modules for EKS node groups, RDS instances, IAM roles, and S3 — replacing manual console-driven changes and eliminating configuration drift across dev, staging, and production.
- Deployed ExternalDNS across all Kubernetes clusters with Route 53 private hosted zones, automating DNS record management for internal services.
- Implemented Okta SSO authentication for Jenkins and ArgoCD, centralizing identity management and eliminating per-service credential management across the platform.
- Established AWS Secrets Manager as the org-wide secrets standard, building the initial secrets structure and access patterns that scaled into the current ESO-based secrets infrastructure.

Senior Application Support Engineer | Sigma Systems | Toronto, CA (Remote) *Apr 2018 – Apr 2019*

- Served as primary technical escalation point for enterprise clients, owning end-to-end troubleshooting and deployment execution.
- Automated deployment workflows with Shell scripting, reducing manual deployment effort by 30% and eliminating a class of human errors.
- Optimized complex SQL queries for production data diagnostics, improving incident resolution speed and data integrity validation.

DevOps / Automation Engineer | TracFone Wireless | Medley, FL *Feb 2013 – Apr 2018*

- Designed and maintained CI/CD pipelines using Jenkins and UrbanCode AnthillPro for a high-volume consumer wireless platform.
- Developed Python and Shell automation scripts, reducing deployment error rate by 40% and standardizing release processes across 10+ application teams.
- Managed IBM WebSphere Application Server environments, improving stability and uptime for production consumer-facing services.

- Mentored junior engineers on DevOps best practices, automation patterns, and release management — growing team capability and reducing dependency on senior staff.

Network & Systems Administrator | Trax USA Corporation | Coconut Grove, FL *Jul 2011 – Jan 2013*

- Managed IT infrastructure including WebLogic application servers, VoIP systems, and network operations. Maintained backup and disaster recovery processes for business continuity.

EDUCATION

Bachelor of Science, Computer Engineering

University of Miami — Coral Gables, FL | December 2010

PROJECTS

Multi-Tenant Cloud ERP — Personal Project

Solo-built multi-tenant cloud ERP (billing, payroll, e-invoicing, and tax compliance) using AI-augmented development with a two-agent handoff pattern (planner → executor). 95 database migrations, 7 architecture decision records. Demonstrates production-grade architectural discipline applied through AI-augmented engineering.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (In Progress)

Certified Kubernetes Administrator (CKA) (In Progress)